

Ephrem Tibebe Mekonnen

Final-Year PhD Candidate in Computer Science

Technological University Dublin

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RESEARCH INTERESTS

Explainable Artificial Intelligence (XAI), Time Series Analysis, Time Series Classification, Machine Learning, Deep Learning, Model Interpretability, Human-Centred AI, Post-hoc Explainability.

EDUCATION

PhD in Computer Science (Expected Thesis Submission: September 2026)	Oct 2022 - Present
Technological University Dublin, Ireland	
Dissertation: Explaining Time Series Classifiers Through Post-hoc XAI Capturing Temporal Dependencies	
Supervisors: Dr. Pierpaolo Dondio and Dr. Luca Longo	
MSc in Computer Science	2019-2021
University of Trento, Italy	
BSc in Computer Science	2013-2017
Addis Ababa University, Ethiopia	

PUBLICATIONS

Journal Articles

- Mekonnen, E.T.,** Longo, L., and Dondio, P. (2025). LOMATCE: Local Model-Agnostic Time Series Classification Explanation. *IEEE Access*.
- Mekonnen, E.T.,** Longo, L., and Dondio, P. (2024). Global Model-Agnostic Rule-Based XAI Method Based on Parameterised Event Primitives for Time Series Classifiers. *Frontiers in Artificial Intelligence*.

Conference Papers

- Mekonnen, E.T.,** Longo, L., Rizzo, L., and Dondio, P. (2026). L2GTX: From Local to Global Time Series Explanations. *Proceedings of the 4th World Conference on Explainable Artificial Intelligence (XAI 2026)*.
- Mekonnen, E.T.,** Longo, L., and Dondio, P. (2025). Explaining Time Series Classifiers Through Post-Hoc XAI Methods Capturing Temporal Dependencies. *Doctoral Consortium, 3rd World Conference on Explainable Artificial Intelligence (XAI 2025)*.
- Mekonnen, E.T.,** Longo, L., and Dondio, P. (2024). Interpreting Black-Box Time Series Classifiers using Parameterised Event Primitives. *Late-Breaking Papers Track, 2nd World Conference on Explainable Artificial Intelligence (XAI 2024)*.
- Mekonnen, E.T.,** Dondio, P., and Longo, L. (2023). Explaining Deep Learning Time Series Classification Models using a Decision Tree-Based Post-Hoc XAI Method. *Late-Breaking Papers Track, 1st World Conference on Explainable Artificial Intelligence (XAI 2023)*.
- Mekonnen, E.T.,** Brutti, A., and Falavigna, D. (2022). End-to-End Low Resource Keyword Spotting Through Character Recognition and Beam Search Re-scoring. *Proceedings of the IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP 2022)*.
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TEACHING EXPERIENCE

Assistant Lecturer	2017–2019
Haramaya University, Ethiopia	

- Delivered undergraduate teaching in Computer Science modules.
 - Prepared lectures, tutorials, laboratory sessions, and assessments.
 - Supervised student practical work and supported academic advising.
 - Assessed coursework, assignments, and examinations.
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RESEARCH EXPERIENCE

Research Intern

Feb 2021 – Oct 2021

Fondazione Bruno Kessler (FBK), Trento, Italy

- Conducted research in speech processing and machine learning.
 - Developed and evaluated deep learning approaches for low-resource keyword spotting.
 - Contributed to a peer-reviewed publication presented at ICASSP 2022.
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TECHNICAL SKILLS

Programming: Python, SQL

Machine Learning: Explainable AI (XAI), Deep Learning, Time Series Analysis

Frameworks: PyTorch, scikit-learn, tsai

Research Tools: LaTeX, Git

OTHER WORK EXPERIENCE

Software Developer

Apr 2022 – Jul 2022

LIFETOUCH Srl, Parma, Italy

- Implemented WebSocket communication between users and the server to support real-time rover monitoring and control.
- Developed RESTful APIs for communication between the server and autonomous rover, handling path instructions and status updates.
- Contributed to a Flutter mobile application for real-time rover tracking and map-based visualisation.

Software Engineering Intern

Jul 2017 – Sep 2017

Ethiopian Airlines, Ethiopia

AWARDS AND SCHOLARSHIPS

MSc Scholarship, University of Trento

2019–2021

Selected Participant, Cornell, Maryland, Max Planck Pre-doctoral Research School

2020

REFEREES

Available upon request.